Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/20/2007
PI Name:	Koplik, Joel Ph.D.		
Project Title:	Molecular Dynamics of Fluid-Solid Systems		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2007 per PI (10/06) Note - grant renewed for 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:	Koplik, Joel		
COI Name (Institution):	Banavar, Jayanth (Pennsylvania State University)		
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Task Description:	The numerical calculations are based on Molecular Dynamics, wherein one integrates the (classical) equations of motion for interacting molecules of various species, based on a prescribed interaction and specified thermodynamic operating conditions and mechanical forces. Within the framework of molecular dynamics, it is straightforward to impose a desired value of gravity or simulate the presence of g-jitter. The objective of this theoretical research is to examine the molecular behavior of fluids in confined geometries, to determine flow boundary conditions, and to investigate non-Newtonian rheology, using molecular dynamics calculations and the statistical mechanics of classical fluids. We consider temporal and spatial scales where continuum fluid mechanics provides little insight. Examples under investigation include (1) phase behavior and spreading of surfactant solutions, (2) solidification, evaporation, coalescence and rupture of non-Newtonian fluid drops and filaments, (3) boundary conditions for dilute gases and other interfaces (4) dynamics of a polymer molecule in solution, and (5) the physics of colloidal hard sphere systems. We intend to carry out a number of calculations involving coexisting liquid, vapor and solid phases under flow conditions, with two common features. First, the systems will be studied at the molecular scale to obtain otherwise unavailable information. Second, the problems to be studied deal with fundamental issues in the physics of fluids and interfaces that are of relevance to the microgravity program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research provides an improved understanding of the microscopic physics underlying macroscopic fluid pheneomena in several areas, which is expected to lead to better design and fabrication in future applications.
Task Progress:	Final Report NASA Grant NNC04GA27G 2003-2007 Molecular Dynamics of Fluid-Solid Systems Joel Koplik and Jayanth R. Banavar During the grant period, a number of topics involving molecular-level studies of interaction fluid-solid systems were considered. 1. non-Newtonian Rheology We investigated several aspects of non-Newtonian filament stretching, with the two aims of relating microscopic simulational information to continuum behavior, and of studying the late stages of extrusion when the fluid interface ruptures. In the first case, the extensional rheology of Newtonian and non-Newtonian fluids was investigated by molecular dynamics simulations of corresponding model liquids in an extending liquid bridge configuration. The fluids were modeled as molecular species and the polymers as chains of monomers bonded by freely-jointed non-linear springs, interacting with atomistic solid walls. We considered three model systems -- a Newtonian liquid made of short chains, a polymer solution consisting of longer chains in a Newtonian solvent, and monodisperse and polydisperse polymer melts made of long chains. In the simulations, a cylinder of liquid was placed between solid end plates, separating at an exponentially increasing rate until the liquid breaks up. A quantitative description of the time evolution of the liquid filament profile and the forces exerted on the end-plates was obtained, which can be directly compared to experiment. In addition, monitoring of the atomic displacements and forces during the simulation yields information on the internal dynamics of the fluid - the velocity and stress fields, and the molecular configurations. The simulations are in good agreement with laboratory data and with the results of macroscopic numerical calculations where available, but provide new and detailed information on the internal dynamics of liquids in extensional flow. In the second case, we presented molecular dynamics computer simulations of filaments of model non-Newtonian liquid stretched in a uniaxial deformation to the point of breaking. The liquid consisted of Lennard-Jones monomers bound into chains of 100 monomers by nonlinear springs, and several different constant velocity and constant strain rate deformations were considered. Generally we observed nonuniform extensions originating in an interplay between the stretching forces and elastic and capillary restoring mechanisms, leading to highly uneven shapes and alternating stretched and unstretched regions of liquid. Except at the fastest pulling speeds, the filaments continued to thin indefinitely and broke only when depleted of molecules, rather than common viscoelastic rupture mechanisms. 2. Wetting and Particle Attachment to Walls We used molecular dynamics simulations to study the behavior of a closely-fitting spherical particle in a cylindrical fluid-filled tube of nanometer diameter. Although the early-time motion of the particle was in agreement with continuum hydrodynamic calculations, we observed, in cases where the surrounding fluid only partially wets the particle and walls, a subsequent attraction of the particle to the wall and various sliding, rolling and stick-slip motions 3. Surfactant-facilitated spreading The spreading of a partially wetting aqueous drop on a very hydrophobic surface can be facilitated by the adsorption of surfactants onto the vapor/liquid and liquid/solid interfaces of the drop. The precise mechanism is unknown, as is the fluid mechanics near the contact line when surfactants are present. A key observation is that while it is easy to reduce the vapor/liquid surface tension with any number of surfactants, only in few cases is it possible to make a non-wetting liquid spread, and therefore the structure of the surfactant molecule is critical. For example, flexible, linear surfactants with polar groups attached to one end of a straight alkyl hydrophobic chain slightly enhance wetting, but "T" shaped trisiloxane surfactants, with polar groups attached at the center of a trisiloxane hydrophobic chain, leads to complete wetting of aqueous drops ("superspreading"). We are exploring the possibility, suggested by several experimental studies, that spreading occurs via a precursor film made of a surfactant bilayer which sandwiches the weakly-wetting solvent. In order to reduce the computational demands, our initial simulations are based on an analog non-polar system rather than a realistic aqueous solvent, with a monatomic solvent, a particle-lattice substrate, and surfactants modeled as bead and chain constructs, with all interactions given by Lennard-Jones potentials. We consider "T" shaped and flexible linear

chain surfactant structures, and interaction strengths selected so that the surfactant is insoluble. We have shown that as a drop spreads the "T" shaped surfactant monolayers on the interfaces can, when the interaction between the solvent and the hydrophile particles is strong enough, assemble into a bilayer, allowing the drop to extend to a thin planar film. In contrast, for a flexible linear chain surfactant there is no coordinated interaction between the monolayers on the two interfaces, due to less efficient intercalation, and little effect on spreading. More sophisticated larger scale simulations have begun, using realistically modeled water and inter-species interactions.

5. Slip at Liquid-Liquid Interfaces

In order to investigate the question of possible slip at a liquid-liquid interface, we have undertaken molecular dynamics simulations of the Couette and Poiseuille flows of two-layered liquid systems with various molecular structures and interactions. In all cases, the average velocity is found to vary continuously across the interface. However, when the total liquid density in the interfacial region drops significantly compared to the bulk values, the tangential velocity varies very rapidly there, and would appear discontinuous at a coarse or continuum resolution. The value of this apparent slip varies linearly with the shear stress at the interface, with a constant of proportionality depending on the nature of both liquids but not the flow configuration. Thus, a version of the Navier boundary condition appears to apply in this situation.

6. Flow on Wetting-Patterned Surfaces

We are investigating the flow of a nano-scale incompressible liquids confined to "chemical channels" on a substrate: a region of solid which is completely wetting to a particular liquid, surrounded by non-wetting solid regions. For the case of a long straight wetting strip, the a uniform strip of liquid is observed in molecular dynamics simulations to develop bulges or pearls, due to a surface-tension instability. The pearls propagate when the liquid moves under a pressure gradient, and combine in pairs due to non-linear wave interactions, but do not obstruct transport. A stability calculation and a simple long-wavelength approximation give qualitatively consistent results, and a paper summarizing this work is in preparation. We have continued with a study of "junction" flows, where the chemical channel has diverging and converging elements. Again, liquid moves just along the channel in the presence of a pressure gradient, and a pearling instability appears but does not obstruct transport. We are now focusing on issues of how to control switching and bifurcation effects at junctions.

Bibliography Type:	Description: (Last Updated: 12/21/2007)
Articles in Peer-reviewed Journals	Rauscher M, Dietrich S, Koplik J. "Shear flow pumping in open micro- and nanofluidic systems." Phys Rev Lett. 2007 Jun 1;98(22):224504. PMID: 17677850 , Jun-2007